Fiscal Year:	FY 2017	**Task Last Updated:**	FY 12/28/2017
PI Name:	Dorny, Jonathan M.S.		
Project Title:	5D Task Analysis Visualization Tool Phase II		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HFBP**:Human Factors & Behavioral Performance (IRP Rev H)		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	93117-3028	**Congressional District:**	24
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	SBIR Phase II
Start Date:	05/04/2017	**End Date:**	05/03/2018
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JSC
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:			
Performance Goal No.:			
Performance Goal Text:			
	The creation of a five-dimensional task analysis visualization (5D-TAV) software tool for Task Analysis and Workload Planning using multi-dimensional visualization will have significant positive impacts on the optimization of human-centered design at NASA. Recent research identified a 40% improvement in task analysis accuracy and efficiency using 3D visualization. Employing enterprise data management innovation, configuration management, and loosely-coupled reusable libraries provides a unified data-driven 5D model portraying total complexity (risk, resources, workload, and duration) and process flow including conditional paths. Critical path and conflicts are accentuated with user controlled sensitivity thresholds for filters, indicators, notifications. The software tool promises increased awareness for human factors, project management, operations personnel, and system designers improving efficiency, accuracy, and understanding. Robotic activity developers and small spacecraft swarming coordinators are provided with an intuitive 3D design representation. These improvements will lead directly to improved system design, optimization of human and		

Task Description:	system allocations, and conservation of resources for long-duration missions including sustainable habitats. The 5D-TAV tool's open architecture integrates open source software to provide the 2D rendering and 3D model views, filters, rotations, and controls necessary for successful task analysis optimization to ensure mission success. POTENTIAL NASA COMMERCIAL APPLICATIONS: The Five-Dimensional Task Analysis Visualization (5D-TAV) will prove valuable for use in NASA Human Factors analysis for long duration missions that include self-sustaining environments with limited resources. The tool can be applied to any complex system development effort that incorporates complex human operations or system design in a collaborative engineering environment. Integration of sensor data will enable mission control be alerted to off-nominal conditions during mission execution and to quickly determine a recovery plan. Robotics will benefit from the tool to design complex routines such as autonomous robots for unassisted maintenance or checking of personnel health, equipment health, and consumable status with automatic assessment of risk, error handling, and resource optimization. New researchers for small spacecraft can utilize the tool for defining and controlling tasks allocated to different spacecraft for coordinated swarming. Additional applications include: - Development of training solutions for high stress environments; - Remote / Autonomous control of equipment and systems; - Emergency preparedness and emergency response; - Integrated workforce planning and program management; - Optimization of field service and sustainment support across enterprises; - Code development performance metrics.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	POTENTIAL NON-NASA COMMERCIAL APPLICATIONS: Existing CPC projects with Army and Navy can all take advantage of the 5D-TAV capabilities for workforce planning, supply chain management, obsolescence control, and condition-based maintenance (CBM) optimization. There are additional opportunities with the Army and Navy to develop service-wide applications to support complex system planning, management, and support requirements. Applications include but are not limited to: - Development of training solutions and measure effectiveness; - Remote / Autonomous control of equipment and systems; - Emergency response planning and training; - Integrated workforce planning and program management; - Optimization of field service and sustainment support across enterprises; - Code development performance metrics; - Task allocation for coordinated autonomous vehicle interaction.
Task Progress:	New project for FY2017. Reporting not required for this SBIR Phase 2 project.
Bibliography Type:	Description: (Last Updated:)